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Automated marine oil spill detection algorithm based on single-image generative adversarial network and YOLO-v8 under small samples

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ABSTRACT

As marine resources and transportation develop, oil spill incidents are increasing, endangering marine ecosystems and human lives. Rapidly and accurately identifying marine oil spill is of utmost importance in protecting marine ecosystems. Marine oil spill detection methods based on deep learning and computer vision have the great potential significantly enhance detection efficiency and accuracy, but their performance is often limited by the scarcity of real oil spill samples, posing a challenging to train a precise detection model. This study introduces a detection method specifically designed for scenarios with limited sample sizes. First, the small sample dataset of marine oil spill taken by Landsat-8 satellite is used as the training set. Then, a single image generative adversarial network (SinGAN) capable of training with a single oil spill image is constructed for expanding the dataset, generating diverse marine oil spill samples with different shapes. Second, a YOLO-v8 model is pretrained via the method of transfer learning and then trained with dataset before and after augmentation separately for real-time and efficient oil spill detection. Experimental results have demonstrated that the YOLO-v8 model, trained on an expanded dataset, exhibits notable enhancements in recall, precision, and average precision, with improvements of 12.3 %, 6.3 %, and 11.3 % respectively, compared to the unexpanded dataset. It reveals that our marine oil spill detection model based on YOLO-v8 exhibits leading or comparable performance in terms of recall, precision, and AP metrics. The data augmentation technique based on SinGAN contributes to the performance of other popular object detection algorithms as well.

1. Introduction

In recent years, with the exploration of marine resources and the increasing number of operating vessels, the recognition and detection of marine oil spill gradually attract attention. Marine oil spill has a serious negative impact on marine biodiversity, habitats of birds and marine mammals, fishery resources and coastal economic activities. Early monitoring and detection of marine oil spill greatly facilitates the identification of the oil spill source. Traditional methods for manually marking and inspecting marine oil spill (Hong et al., 2022) relies on the knowledge and experience of researchers, making it time-consuming, labor-intensive, and susceptible to missing and erring examinations.

With the rise of artificial intelligence, the detection method of marine oil spill is being innovated. Through the application of artificial intelligence technology, some limitations and challenges in traditional methods can be effectively overcome. There are many machine learning algorithms, such as Support Vector Machine (SVM) (Matkan et al., 2013), Random Forest (RF) (Ozigis et al., 2020), and k-means algorithm (Chen et al., 2022), that can ameliorate the reliability of ocean oil spill detection. Recently, neural network approaches have been introduced to enhance the detection performance of marine oil spill (Mohammadiun et al., 2021). Due to the structure of multiple hidden layers in deep neural networks and the capacity of learning hierarchical features from high-dimensional raw data, the detection accuracy of deep learning models is typically higher when compared to traditional techniques (Yang et al., 2022). For example, Convolutional Neural Networks (CNNs) have been extensively used for marine oil spill detection tasks (Li et al., 2022a; Dehghani-Dehcheshmeh et al., 2023) and have shown considerable performance. Also, other researchers introduced the variants of CNN such as U-Net (Chen and Wang, 2022), AlexNet (Wang et al., 2021), Fast-RCNN (Huang et al., 2022) to identify marine oil spill.

Despite the advantage of the DL-based approach for marine oil spill detection described above, some challenges remain. One of the main challenges is to provide an appropriate amount of data to train the

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network. However, the real cases of marine oil spill are relatively limited, which leads to a lack of available data. The establishment of a high quality marine oil spill data set is usually a time-consuming procedure that requires considerable effort and cost, especially when the network requires large number of samples. There are many methods of data augmentation, such as clipping, flipping, scaling and Gaussian noise. However, these augmentations are usually being implemented with simple transformation and in general can not improve the training quality of the DL model. Generative adversarial networks (GANs) offer an alternative when facing the issue of a lack of training samples by generates composite data samples with diversity and fidelity. The powerful generation capability of GANs enables the provision of a greater number of training samples for model training. This data augmentation technique effectively enhances the model's generalization performance, expands the data coverage, and mitigates the challenges posed by limited data availability. Especially, a class of GANs called Single Image Generative Adversarial Networks (SinGAN) is proposed that can generate diversified, high-quality images from a single image (Shaham et al., 2019).

Another challenge of DL-based approach for object detection is to improve the real-time performance and the accuracy of detection. YOLO-v8 (You Only Look Once version 8), a target detection algorithm based on deep learning, has demonstrated its high performance and high accuracy compared to many other traditional CNN models (Fang et al., 2022), solidifying its position as a powerful solution in the field of object detection. YOLO-v8 stands out by enabling end-to-end training without the need for additional preprocessing or post-processing steps. By treating the target detection task as a single regression problem and simultaneously predicting the location and category of the targets, YOLO-v8 achieves a streamlined and efficient model design.

This work aims to improve the marine oil spill detection process by using a YOLO-v8 framework for detection and a SinGAN framework for high-quality data augmentation. The main contributions of this paper are given as follows:

1. Before this work, we collected a dataset comprising 5000 OLI images from Landsat-8. The oil spills in these OLI images were labeled through expert visual interpretation and auxiliary environmental information, resulting in a total of 772 ground-truth oil spill samples.
2. A SinGAN framework that can be trained from a single image is utilized to capture the multi-scale features and texture information of the oil spill, allowing it to create synthetic images which exhibit similar to the input image but possess diversity.
3. We employ the new SEID metric and manual assistance to individually evaluate the generated samples, and utilize these samples to expand the marine oil spill dataset.
4. A latest YOLO-v8 detection framework is trained on original and generated data to achieve a relatively high level of detection accuracy. The use of transfer learning not only expedites the learning process but also reduces the data requirements for the specific task.
5. Compared to other popular detection models, our YOLO-v8-based model not only showcases a leading or comparable performance in terms of evaluation metrics but also excels notably in the processing speed of individual images. The data augmentation technique based on SinGAN contributes to the performance of other popular detection algorithms as well.

The rest of this paper is structured as follows. Section 2 introduces some related works about the marine oil spill detection and the GAN-based data augmentation. Section 3 introduces the information of our dataset and proposes a marine oil spill detection framework based on SinGAN and YOLO-v8. Our experimental results, along with a comprehensive discussion on the detection outcome, are presented in Section 4. Section 5 draws a conclusion of our work and gives some suggestions for future research.

2. Related works

2.1. Marine oil spill detection

Marine oil spill detection can be divided into two types: traditional approaches and DL-based approaches (Vasconcelos et al., 2023). Traditional methods usually use manual-designed feature extraction and classification algorithms to identify marine oil spill. For example, Xu et al. (2017) utilized the OTSU algorithm in oil spill observation, and employed the assumption that images under detection solely contain background image and foreground image and accomplished image segmentation by threshold computation. Li et al. (2022b) used SVM to find out the effective observation location of marine waves and applied FCM algorithm to identify the marine waves and oil films. Li et al. (2022c) proposed a method that combines a SVM with histogram of gradient (HOG) features to improve the exactness of identifying low-altitude off-coast oil spill, enabling all-weather observation in off-coast waters.

Manually designed feature extraction and classification algorithms rely on the knowledge and experience of domain experts, and there is a certain degree of subjectivity and limitations in feature selection and design. This dependence on human expertise can lead to biases and inconsistencies in selected features (Zhang et al., 2023). For complex oil spill images, especially in the case of multispectral oil spill images, manually designed feature extraction algorithms may not be able to fully express the useful information in the image (Liu et al., 2022), resulting in limited recognition performance.

In contrast, machine learning approaches, particularly deep learning, offer an alternative by automatically learning features directly from the data, which can effectively overcome the limitations of manually designed approaches and achieve higher accuracy and generalization performance in oil spill recognition tasks. Dasari et al. (2022) present an approach that employs Supervised Support Vector Machines (SSVM) and Neural Network (NN) to validate the mapping of oil spill detection. The findings exhibit a robust concurrence between the classified images derived from SVM and neural network (NN) classifiers and the oil spill mapping outcomes. Zhai et al. (2022) introduce a novel approach called Dual Attention Encoding network (DAENet) that offers a more feasible and effective approach to accurately identify and monitor marine oil spill areas. Ma et al. (2023) employed the DeepLabv3+ model to detect and identify marine oil spill and proved its superiority to traditional SVM and FCN algorithms. The DeepLabv3+ model is an advanced DL architecture commonly used for semantic segmentation tasks, including the detection and delineation of objects in images. Hasimoto-Beltran et al. (2023) put forward a multi-channel deep neural network model based on the combination of ResNet and U-Net architectures, whose intention is to improve prior DNN oil spills detection models, via taking into consideration multi-channel input, complex oil spill shapes under different scales and in different evolution period. Similarly, Wang et al. (2023) have demonstrated the utilization of ResNet-50 as the underlying framework in DeepLabv3+, coupled with SVM for classifying, enabling swift and precise identification of oil spill. Yu et al. (2018) propose an automatic oil spill segmentation method based on Adversarial f-Divergence Learning which helps alleviate the requirement for an extensive amount of labeled data to train the deep network parameters.

While deep learning methods have shown great potential in marine oil spill detection, they often require a large number of diverse and well-annotated oil spill images for model training. However, such annotated data sets are scarce in the marine oil spill detection scenario. Hence, a marine oil spill generation images approach must be proposed to augment the training of DL models.

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2.2. Generate adversarial networks

2.2.1. Principle of GANs

The concept of GANs was first introduced by Ian J. Goodfellow et al. (2014) in 2014, and since then, it has garnered significant attention from researchers worldwide. GANs have become a popular research area, prompting researchers to delve deeper into their potential applications and explore their capabilities extensively. In GANs, there are two pivotal components: the generator and the discriminator. The generator takes a random noise as input and generates fake images, while the discriminator's role is to distinguish between real and fake images. These two components are trained concurrently in a mini and max game. The used setup is shown in Fig. 1.

The loss function of the generator is designed based on the scalar export from the discriminator i.e. the probability that the generated sample is judged to be a real sample. The goal of the generator is maximizing this probability so that the discriminator can not accurately differentiate between the synthetic sample and the real sample. In the standard GANs, the generator's loss function is determined as the negative logarithm of the probability that the generated sample is assessed to be an authentic sample, that is:

$$\mathscr{L}_G = -\mathbb{E}_{z\sim p_z(z)}[log(D(G(z)))] \tag{1}$$

The loss function of the discriminator is designed to measure the classification accuracy to the real sample x and the generated sample $G(z)$. The intention of the discriminator is to optimize the loss function with the aim of improving its ability to distinguish real samples and generated samples, that is:

$$\mathscr{L}_D = \mathbb{E}_{z\sim p_z(z)}\{log[1-D(G(z))]\} - \mathbb{E}_{x\sim p_{data}(x)}\{log[D(x)]\} \tag{2}$$

The objective function of GANs can be defined as the sum (or difference) of the generator loss function and the discriminator loss function, where the generator loss function takes a negative value, that is:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim P_{data}(x)}[logD(x)] + \mathbb{E}_{z\sim p_z(z)}[log(1-D(G(z)))] \tag{3}$$

Where x is the real sample, $P_{data}(x)$ is the distribution of the real sample, z is the input random vector of the generator, $P_z(z)$ is the distribution of the input random vector, $G(z)$ is the image created by the generator, and $D(x)$ is the possibility that the discriminator gives the input x to be the real sample. The objective function reflects the antagonistic relationship between the discriminator and the generator.

2.2.2. Improvements of GANs models

After the emergence of GANs, its method showed promising outcomes in image generation and demonstrated potential for advancement and application across various domains. Researchers have made valuable contributions in addressing challenges associated with the initial version of GANs. The improvements of GAN models can be classified into three aspects: the architecture, the loss function and the training strategy. Specifically, GANs can be categorized into different types, as shown in Fig. 2.

One of the most famous models in the field of GANs is the Deep Convolutional GAN (DCGAN) model (Radford et al., 2015), which has gained significant recognition in the early stages of research. Different GAN such as CGAN (Mirza and Osindero, 2014), ACGAN (Odena et al., 2017), StyleGAN (Karras et al., 2019) are proposed to enhance the control over the generated results. Zhang et al. introduced self-attention mechanism in generative adversarial networks called SAGAN (Zhang et al., 2019), which enhances the consistency and realism of generated images, resulting in improved image synthesis quality. The BigGAN Brock et al. (2018) model enhance the ability to learn complex features by increasing the capacity and depth of the network. Deep networks progressively extract and combine higher-level features, resulting in more detailed and realistic generated images. UNet-GAN (Schonfeld et al., 2020) combines the U-Net and GAN frameworks for image synthesis and segmentation tasks. Some methods have been proposed for solving the training stability problem by designing an improved loss function such as LSGAN (Mao et al., 2017), RaGAN (Jolicoeur-Martineau, 2018), WGAN (Arjovsky et al., 2017), and WGAN-GP (Gulrajani et al., 2017). Also, there are some other variants of GAN that are improved from training strategies or normalization technique, such as PGGAN (Karras et al., 2017), SNGAN (Miyato et al., 2018), CycleGAN (Zhu et al., 2017). Among various generative models, SinGAN (Shaham et al., 2019) requires only one image for training and can generate novel target shapes and constructions on the foundation of maintaining the original image overall similarity. This makes it possible for SinGAN to carry out image synthesis and data augmentation without large-scale labeled data.

2.2.3. GAN-based data augmentation

Basic image transformations such as translation, cropping, and rotation have a tendency to produce highly correlated samples. These transformations may not adequately capture variations arising from different imaging protocols or sequences. Improper implementation of these transformations may result in ineffective model improvements or

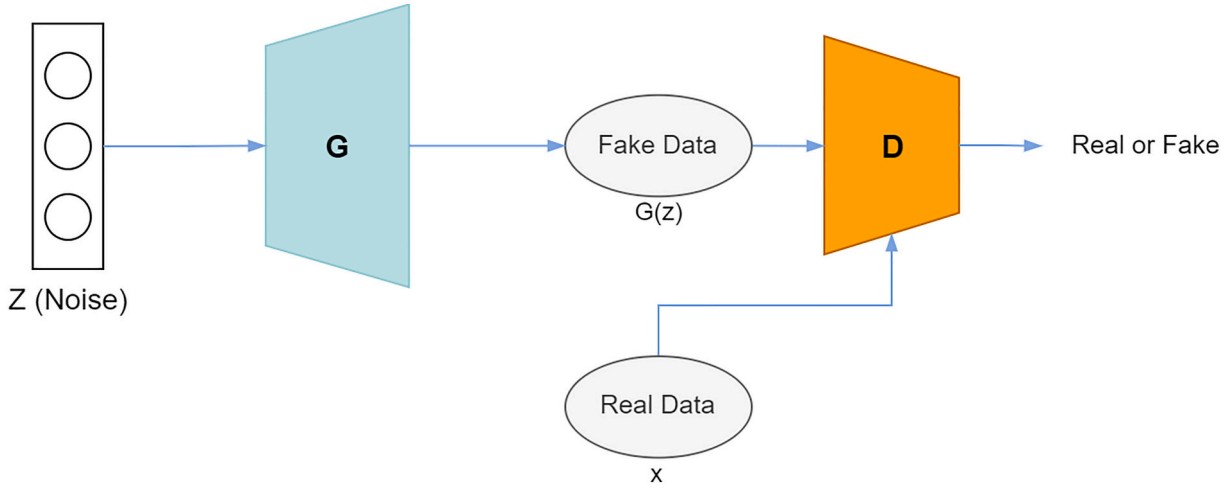


Fig. 1. The generator converts random noise into a sample $G(z)$ that is similar to the real data. As the training goes on, the generator gradually acquires proficiency in generating more realistic samples to deceive the discriminator. Discriminator is aimed to differentiate the synthetic sample $G(z)$ from the true sample x as accurately as possible, so as to improve its discriminant ability.

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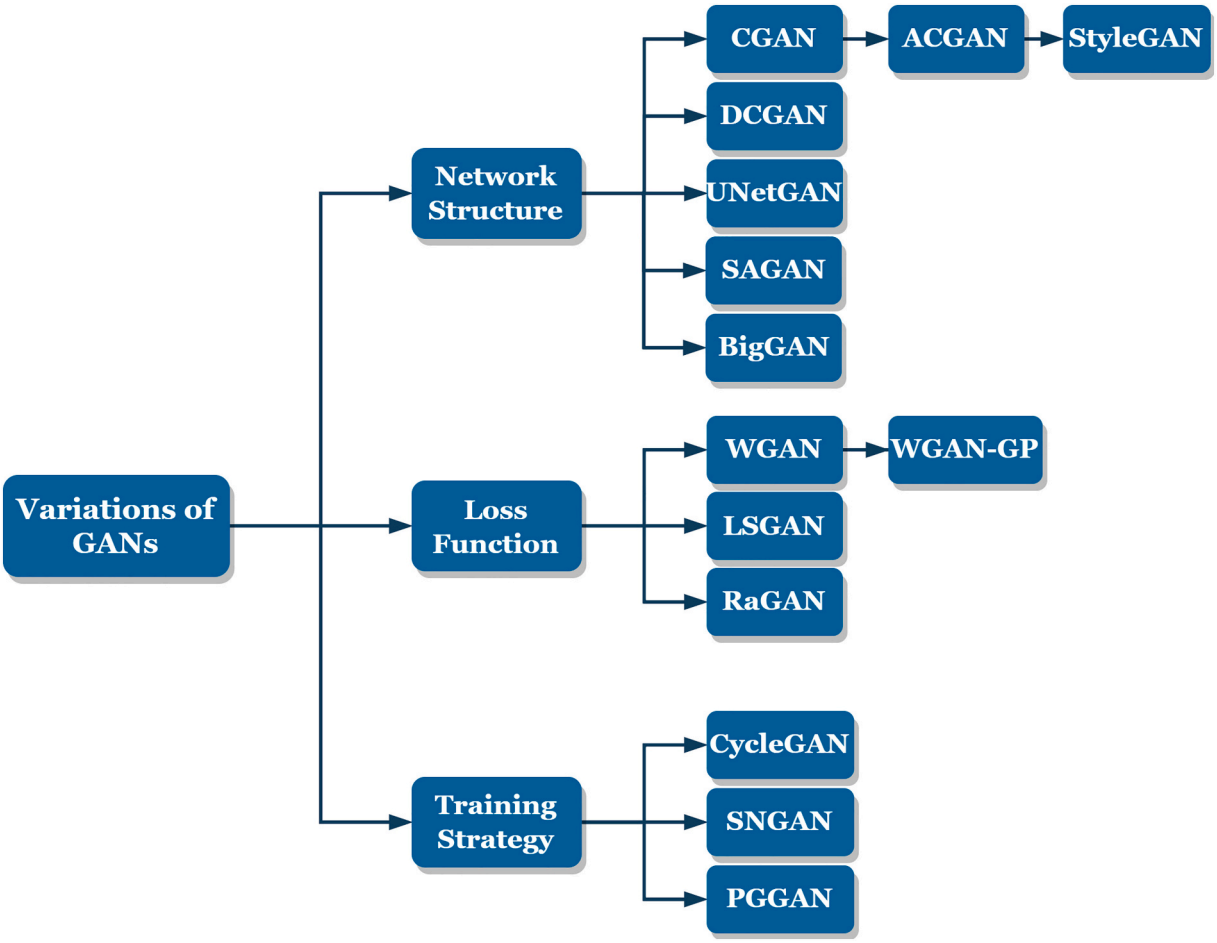


Fig. 2. Famous Variations of GANs.

limited generalization, and in some cases, even lead to overfitting (Han et al., 2023). Typically, these augmentation techniques are applied individually to each image, which can be insufficient for effectively capturing the variations or invariant features present in the rest of the training data. The emergence of Generative Adversarial Networks, however, has significantly contributed to the advancement of data augmentation techniques. GANs are able to aid in generating additional data and expanding the dataset in the situation of data insufficiency.

Zhang et al. (2020) use Generative Adversarial Networks to capture the relationship between noise distributions and actual instantly vibration data from machinery and create fake data to Improve the imbalance and sparsity of machinery fault diagnosis. Li et al. (2021) employed the Multiscale CGAN model to investigate the problem of training an oil spill detection model with small data. In Zhuo and Ge (2020), the author proposed a data augmentation method which combines Generative Adversarial Networks (GAN) with Gaussian Discriminant Analysis (GDA) as an aid to enhance fault classification accuracy. To expand the original small sample dataset, Xu and Liu (2022) employed the DCGAN model to generate additional road crack images and then used a Vgg16 model for road crack detection. In Sun et al. (2023), a data augmentation approach based on ACGAN and Wasserstein GAN for three-phase full-bridge inverter (TFI) fault diagnosis has been proposed. To accomplish the purpose of data augmentation, the generator is used to extract the sample distribution characteristics from a limited number of fault mode samples and synthesize a multitude of fault samples belonging to specific categories. GAN can not only be used for image data augmentation, In the paper by Fan et al. (2022), a credit debt swap sequence generation algorithm is proposed. This algorithm, utilizing a SeqGAN, effectively expands the size of the sample and enhances the performance of the risk evaluation model. In general, GANs' application in data augmentation provides a means to generate more diverse and rich data samples, which can enhance the performance, robustness, and generalization ability of deep learning models.

3. Proposed method

3.1. Overview

Fig. 3 illustrates the overall of the proposed approach on marine oil spill under small samples, including three parts i.e. data acquisition, data augmentation and oil spill detection. First, optical imagery on the ocean surface from Landsat-8 OLI are collected and manually annotated by spatial analysis and spectral diagnosis. Then, utilizing a Single Image Generative Adversarial Network (SinGAN) to train on a single ocean oil spill images in turn and generate high-quality images. This approach effectively expands the original small sample dataset by incorporating the generated samples, thereby enhancing its diversity. SinGAN is trained on a single image, which effectively solves the problems of insufficient training samples and difficulty in convergence in traditional GANs. Subsequently, the original small sample data set and the expanded data set are separately applied to train the YOLO-v8 model, allowing for a comprehensive evaluation of the detection performance.

3.2. Data acquisition and processing

Fig. 4(a) provides geographical extent of the coverage area of our data source in the North South China Sea (NSCS), spanning 105°E to 124°E longitude and 13°N to 26°N latitude. The image utilized in this

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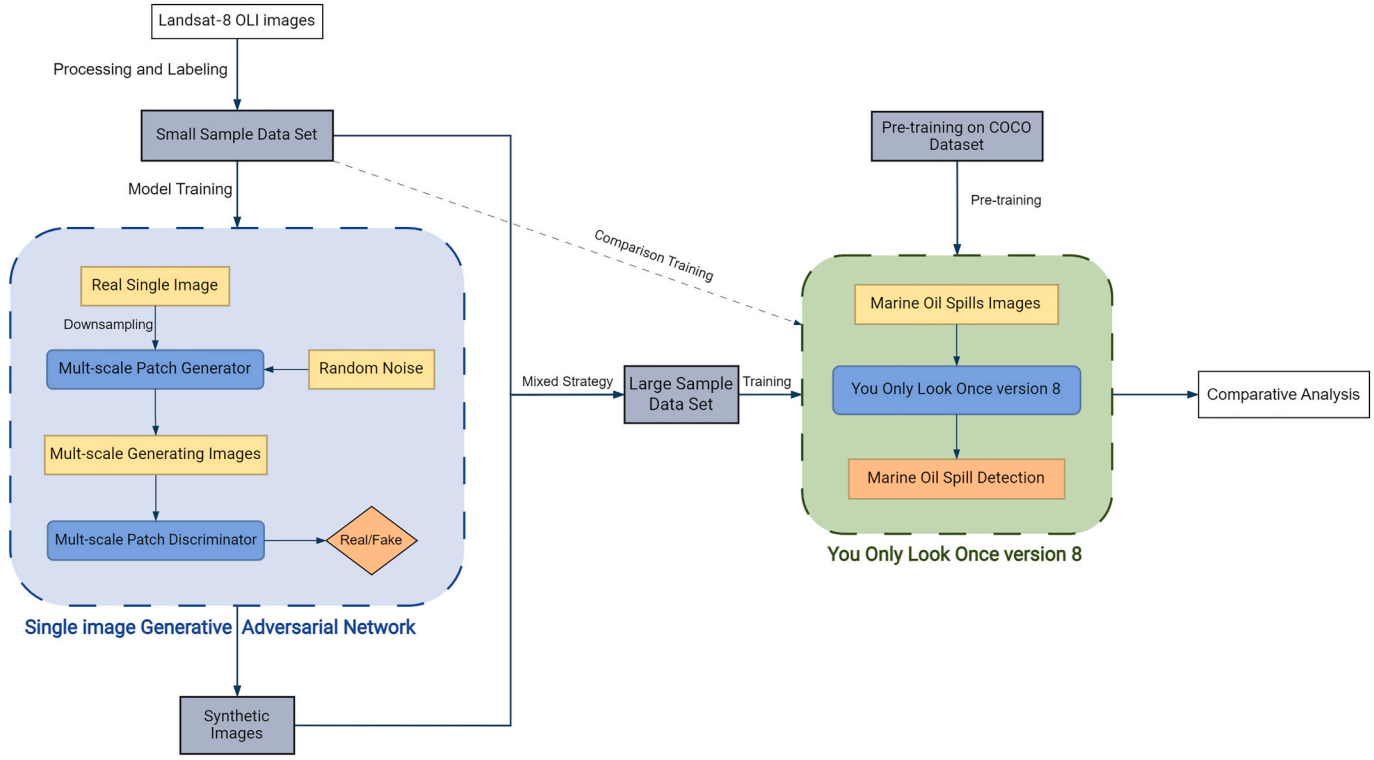


Fig. 3. The proposed method of marine oil spill detection method based on SinGAN and YOLO-v8.

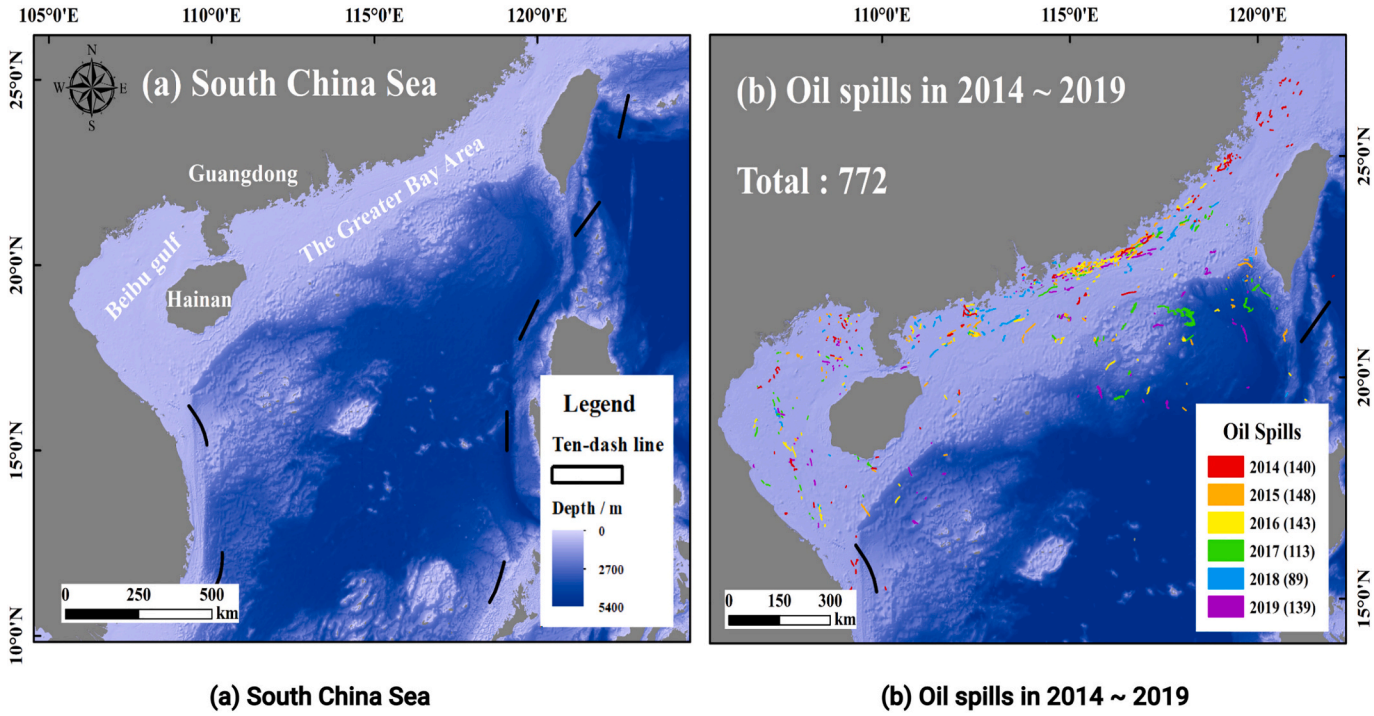


Fig. 4. Map of the study area.

paper has been sourced from the Landsat-8 satellite spanning the period of 2014 to 2019. From the United States Geological Survey (USGS),[1] we downloaded Landsat-8 OLI images having multi-wavelength bands and geospatial 30 m resolution. Table 1 presents some details of the Landsat-8 OLI original images utilized in our work. The collected data of Landsat-8 OLI was publicly accessible throughout the course of the research.

Oil spill in the marine generally shows a series of different features to the clear seawater. In optical imagery, oil slicks always present a dark or light and dark stripes on the sea surface, which create spatial contrasts

[1] https://earthexplorer.usgs.gov/.

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Table 1
Specification.

Specification	Value
Resolution	30 m
Shooting Interval	16 days
Wavelength	430 nm–2290 nm
Running Time	2013-now
Bands	8
Origin	USGS

between oil and background seawater under certain condition (Blondeau-Patissier et al., 2023). The spatial anomalies mainly due to the coverage of oil slicks, which dampen sea surface capillary and short gravity waves. Traditional algorithms for marine oil spill detection often rely on Synthetic Aperture Radar (SAR) imagery data for analysis and detection. Compared to SAR imagery, OLI provide a greater abundance of spectral information, making them more advantageous for visual interpretation and assessment of oil spills. In a cloud-free imagery, it's easy to distinguish thick oil slicks from the seawater by visual interpretation (Dong et al., 2022). However, thin oil spill (thickness $< 0.09\mu m$) might not be able to influence the sea surface waves, leading to the weak or no contrast and the visual interpretation failure. Meanwhile, the weak sunglint might also lead to this failure. Based on the aforementioned prior knowledge, we have identified the marine oil spill depicted in Fig. 4(b) through visual interpretation. Despite the limit, we still can catch most small oil slicks in the ocean through the above theory. Specifically, we firstly obtained the Landsat-8 surface reflectence imagery; then the stretched RGB true color imagery was used to manual-checked; secondly, among all slicks, the oil look-alikes would carry out a check round for further confirm (more detail can be found in Hong et al. (2022)); thirdly, exclude the oil look-alikes and determine the oil spills database; finally, boxed out the shape of oil slicks on LabelImg (Tzutalin, Labelimg, 2015).

It is advantageous to utilize smaller-sized images. To achieve this, we partition the large-size image into multiple smaller images, with each image's size being less than 512×512. After removing images showing anomalies, 772 images containing oil spill are gathered. Out of these images, 80 % has been allocated as the training sets, while the remaining images for the test sets. Some of the samples are illustrated in Fig. 5. Also, we add the cropped images that do not contain oil spill as background images to the dataset for training, which assists the model learn in differentiating between oil spill and backgrounds, reducing the occurrence of false positives.

Labels for image should be exported to YOLO schema with each image having a corresponding *.txt file. If there are no oil spill in an image, no *.txt file is required. Each object in the *.txt file ought to be represented in the $<$ *class* X Y W H $>$ schema, with one row allocated for each. All coordinate parameters need to be normalized to a range of 0 to 1, and the integer representing the categories should start with 0. Given that our focus is solely on detecting marine oil spill, we assign the number 0 as the first value for each line in the *.txt file, representing this specific category. The method of determining the parameters *X Y W H* is illustrated in Fig. 6. The label file corresponding to the image contains 1 marine oil spill.

To annotate the images in the original dataset and augmented dataset, we utilized a powerful tool called LabelImg (Tzutalin, Labelimg, 2015). LabelImg allows us to draw bounding boxes and keypoints on the images and save to YOLO data format, facilitating subsequent experiments on detecting oil spill.

3.3. Single image generative adversarial network

3.3.1. Multi-scale architecture

SinGAN utilizes a pyramid of GANs, implementing a progressive training and inference approach that operates from coarse to fine scales. Instead of utilizing whole image samples from a database, this approach employs training samples in the form of patches extracted from a single image, which avoids the problem of lack of training samples in traditional GAN data augmentation. The generative framework is illustrated in Fig. 7.

In the multi-scale architecture, $\{G_0, \ldots, G_N\}$ form a network pyramid of generators, trained with x: $\{x_0, \ldots, x_N\}$ (imagery pyramid). Each image x_i is obtained by downsampling the base image x using a downsampling factor r^n, with $r > 1$. Downsampling is a process of reducing the resolution of an image, capturing coarser structures and features. Except for the coarsest level, which is only for generating, the input to

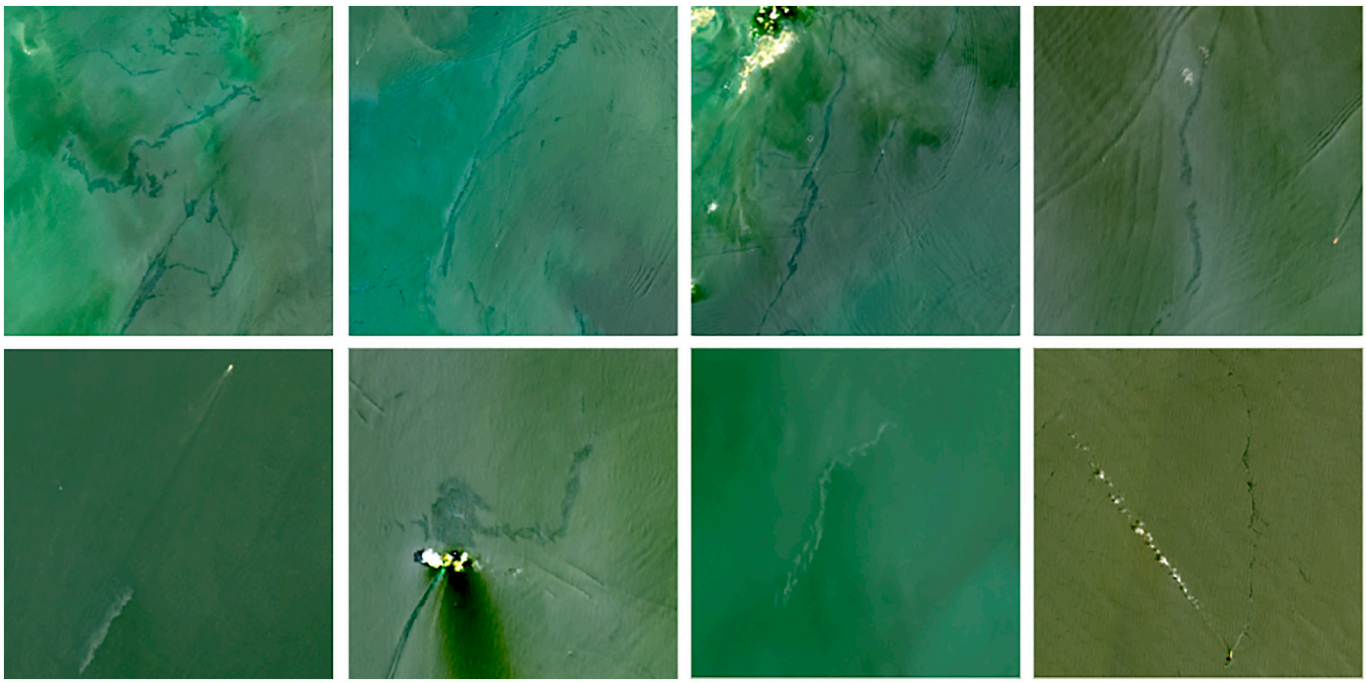


Fig. 5. Some of the marine oil spill samples.

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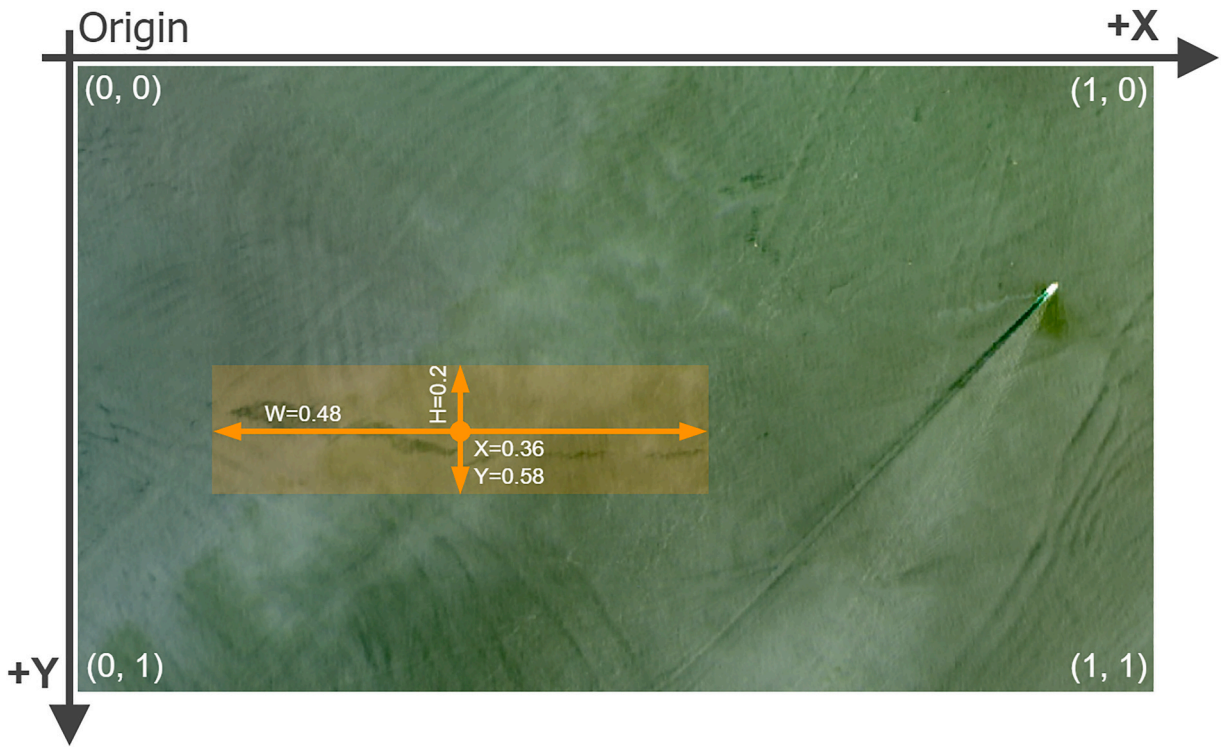


Fig. 6. Determining the parameters *X Y W H*.

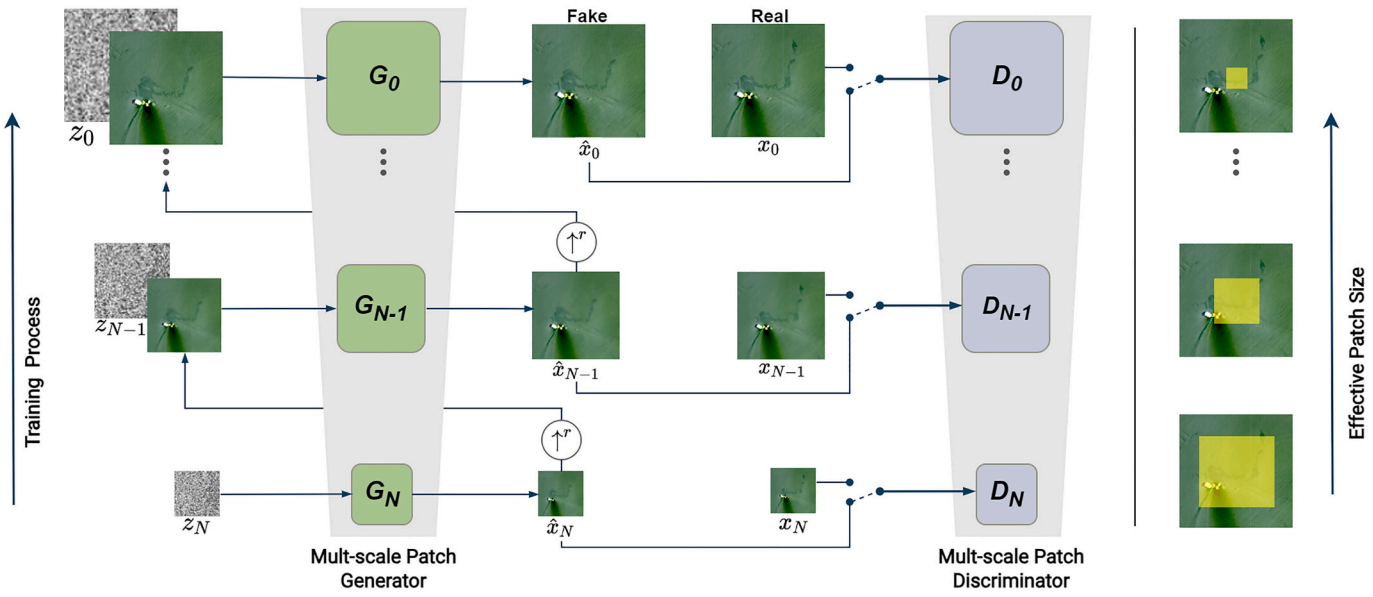


Fig. 7. The multi-scale pipeline of SinGAN for marine oil spill images augmentation.

N^{th} scale generator G_N consists of a randomly generated image z_N and the upsampled generated image from the former scale, $\hat{x}_N$, adjusted to the current resolution. The generators G_N are tasked with deceiving the associated discriminator D_N, whose goal is to differentiate between patches in the synthetic samples and patches in x_N.

Image samples are sequentially generated, beginning from the least fine scale and progressing through all generators until the most delicate scale, accompanied by noise injection at each scale. Each discriminator and generator has a common receptive field, with progressively smaller effective patch size throughout the generation process. At the initial scale, the effective patch size is roughly half of the image's height. G_N converts spatially correlated Gaussian white noise z_N into an image sample $\hat{x}_N$,

$$\hat{x}_N = G_N(z_N). \tag{4}$$

When $n < N$, every generator G_n incorporates an upsampled form of the sample from the previous scale, alongside the spatial noise z_n, that is,

$$\hat{x}_n = G_n(z_n, (\hat{x}_{n+1})\uparrow^r). \tag{5}$$

Fig. 8 showcases the architecture of each generator at different scales. The random noise z_n and $(\hat{x}_{n+1})\uparrow^r$ are merged and passed through a series of convolutional layers. G_n can be understood as executing the

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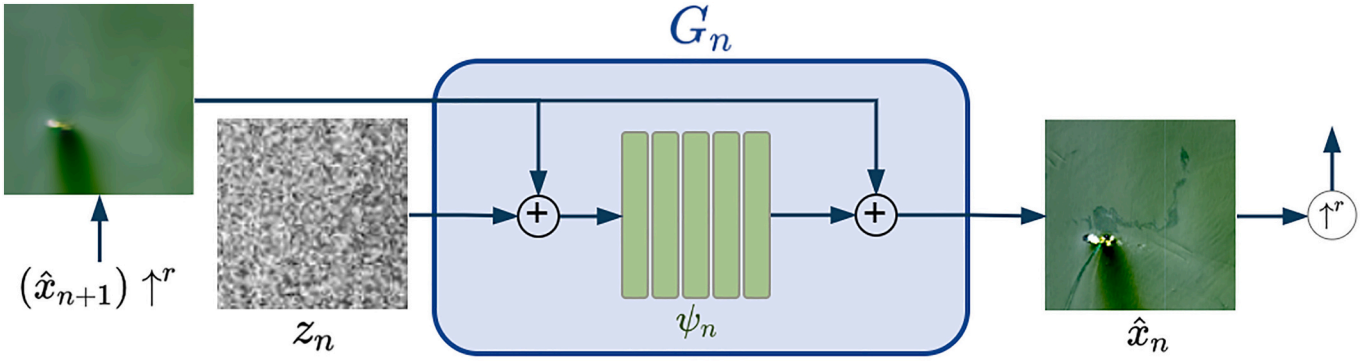


Fig. 8. The architecture of each generator at different scales. The random noise z_n and $(\hat{x}_{n+1})\uparrow^r$ are linearly superimposed and fed into ψ_n. The result of ψ_n is added back to $(\hat{x}_{n+1})\uparrow^r$ as the output of G_n.

following operations:

$$\hat{x}_n = (\hat{x}_{n+1})\uparrow^r + \psi_n(z_n + (\hat{x}_{n+1})\uparrow^r). \tag{6}$$

where ψ_n consists of 5 convolution blocks of the architecture Conv(3 × 3)-BN-LeakyReLU proposed in Ioffe and Szegedy (2015). The number of kernels per block is typically determined by the formula $32 \times 2^{\lfloor \frac{N-n}{4} \rfloor}$, that means we double the number of kernels every 4 scales. The backbone of G_n is similar to ψ_n of D_n.

3.3.2. Loss function

The loss function for the nth scale GAN includes an adversarial loss and a reconstruction loss. That is,

$$\min_{G_n} \max_{D_n} \mathscr{L}_{adv}(G_n, D_n) + \alpha \mathscr{L}_{rec}(G_n). \tag{7}$$

Similar to most GAN, the $\mathscr{L}_{adv}$ ought to measure the distribution of patches between x_n and $\hat{x}_n$. To improve training stability, we adopted the Wasserstein distance used in WGAN-GP (Gulrajani et al., 2017), as the loss function for the generator and discriminator. By incorporating a gradient penalty term, WGAN-GP loss reduces the need for manual learning rate adjustment, resulting in greater flexibility in choosing the learning rate. Additionally, the gradient penalty constraint in WGAN-GP promotes smoother and more manageable training, effectively addressing problems related to mode collapse and training instability. We set the batch size to 1, which means the loss function is defined to consider the entire image as input, instead of dividing the image into smaller random crops and calculating the loss separately for each crop, which enables the network to learn boundaries.

In order to maintain the similarity between the generated oil spill image and the original input image across different scales, we introduce a reconstruction loss item $\mathscr{L}_{rec}$ to accurately preserve the content and structure of the original image at each scale in the generated image. We hope that there is a specific set of noise map so that the generated image $\hat{x}_n$ can be equivalent to x_n. For $n < N$, we assign the specific noise z_n as 0. At the coarsest scale, we define $z_n = z^*$ as a fixed noise map that drawn once and kept unchanged throughout the training process. $\hat{x}_n$ denotes the images generated at various scales using these specific noise maps. Thus, the $\mathscr{L}_{rec}$ can be defined by the following formula.

$$\mathscr{L}_{rec} = \begin{cases} \left\| G_n\left(0, (\hat{x}^{\text{rec}}_{n+1})\uparrow^r\right) - x_n \right\|^2, & when\ n < N \\ \left\| G_N(z^*) - x_N \right\|^2. & for\ n = N \end{cases} \tag{8}$$

3.3.3. Evaluation of generated images

The widely used evaluation metric for GANs is the Frechet Inception Distance (FID) (Heusel et al., 2017). It quantifies the difference between the deep feature distributions of generated images and real images, providing a measure of their similarity or dissimilarity. However, in each training process, we are constrained by only having a single real image available. Also, we are not interested in evaluating the entire set of generated images as a whole. Instead, our primary goal is to individually assess each generated image and reserve those of high quality, in order to construct a diverse and extensive mixed marine oil spill dataset. Therefore, we propose the Single-image Evaluation Inception Distance (SEID) metric. Initially, the Inception model is used to extract feature maps from the single real image and generated image, and flatten these maps into multiple one-dimensional feature vectors. These feature vectors capture the high-level visual characteristics of the image. Next, Calculating the mean vector and covariance matrix of the feature vectors from the single reference image. The SEID is calculated using the following formula,

$$SEID = \| \mu_1 - \mu_2 \| + Tr\left(\Sigma_1 + \Sigma_2 - 2\sqrt{\Sigma_1 \cdot \Sigma_2}\right). \tag{9}$$

Where $\| \mu_1 - \mu_2 \|$ is the Euclidean distance between two means, representing the straight-line distance of two eigenvectors. The second part $Tr\left(\Sigma_1 + \Sigma_2 - 2\sqrt{\Sigma_1 \cdot \Sigma_2}\right)$ is the covariance matrix adjustment term, taking into account the difference in eigenvector distribution. Marine oil spill are considered as sparse targets. Comparing the generated oil spill samples with real samples, there is minimal overall difference, typically with an Euclidean distance of less than 1. As for Fréchet distance, squaring the Euclidean distance would excessively amplify the influence of the covariance matrix adjustment term. Thus, the SEID computation no longer squares the Euclidean distance between the means, enhancing the importance of local differences in the similarity metric calculation and emphasizing the variations among the samples at a local level. We hope that SEID can better reflect the local differences between sparse target samples without being masked by the overall similarity. We employ the SEID metric and manual assistance to individually evaluate a substantial quantity of generated samples, and utilize these samples to expand the marine oil spill dataset.

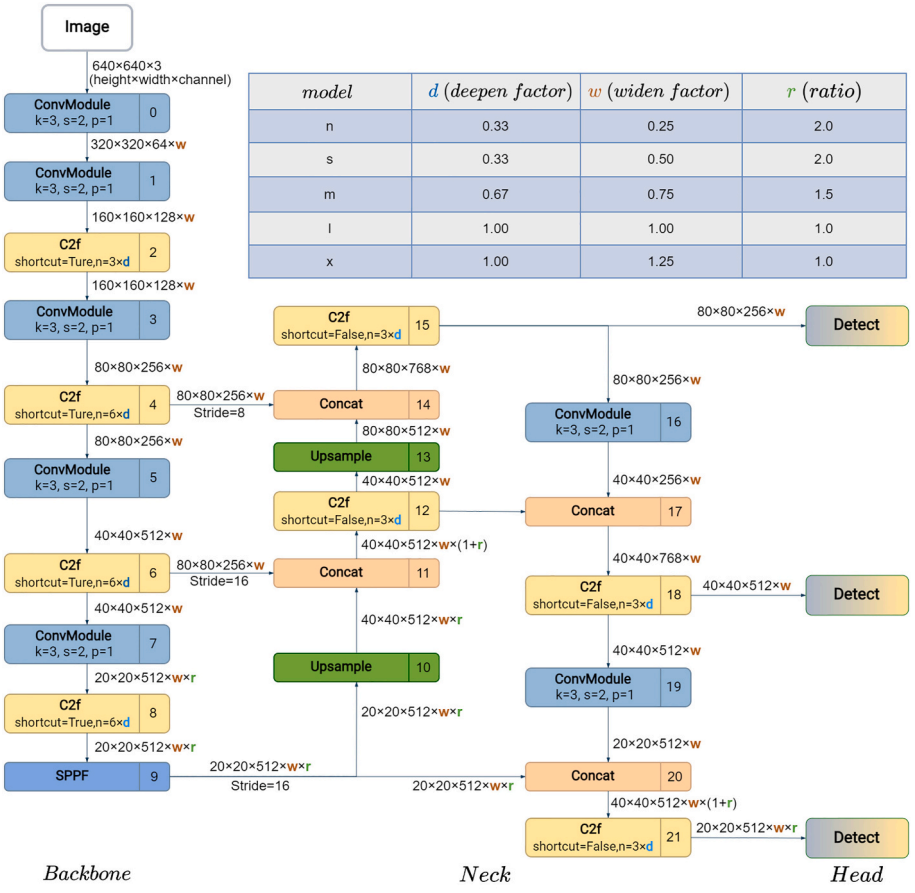
3.4. You only look once version 8

YOLO-v8 (Jocher et al., 2023), developed by Ultralytics, was introduced as an evolution of the YOLO-v5 architecture. YOLO-v8 expanded its offerings to include five versions: YOLO-v8n (nano), YOLO-v8s (small), YOLO-v8m (medium), YOLO-v8l (large), and YOLO-v8x (extra large).

3.4.1. Architecture

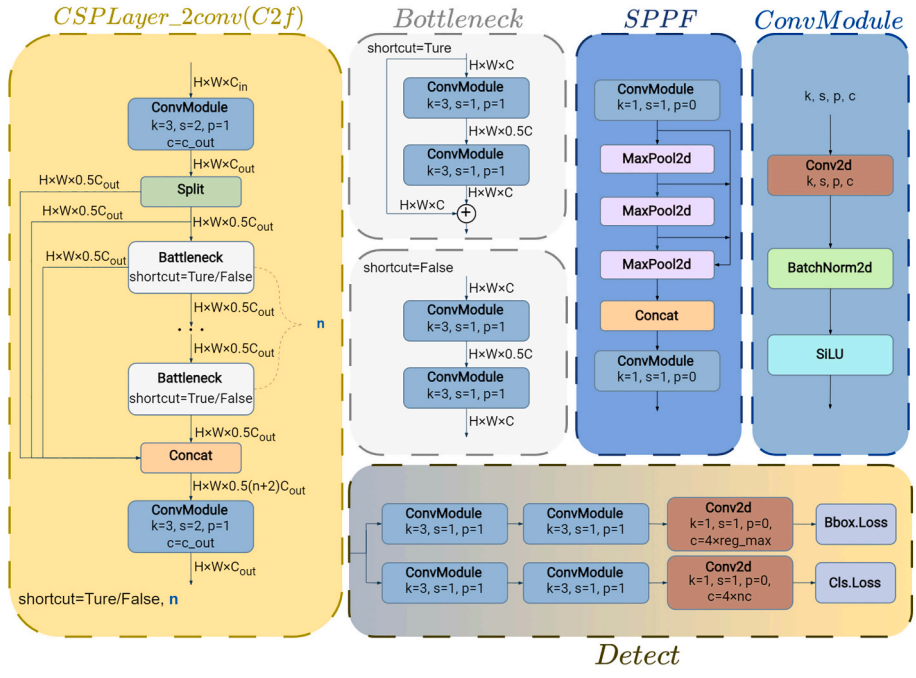
Fig. 9 illustrates the architecture and network specifics of YOLO-v8. While YOLO-v8 uses a similar backbone as YOLO-v5, there are modifications made to the CSPLayer, which is now referred to as the C2f (cross-stage partial bottleneck with two convolutions) module. The C2f module can effectively integrate contextual information with the features of

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model	*d (deepen factor)*	*w (widen factor)*	*r (ratio)*
n	0.33	0.25	2.0
s	0.33	0.50	2.0
m	0.67	0.75	1.5
l	1.00	1.00	1.0
x	1.00	1.25	1.0

(a) Yolov8 architecture



(b) Details

Fig. 9. Illustration of (a) YOLO-v8 architecture, (b) details of the network. The design incorporates an adapted CSP Darknet53 framework. The CSPLayer from YOLO-v5 is replaced by the C2f module. To accelerate computation, an SPPF module is employed to pool features to be a standardized map. BN layer and SiLU structure are applied to each convolution module. The head of the network is decoupled to independently handle object detection, regression, and classification scenes. The above framework builds upon reference (Contributors, 2023).

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higher level to enhance detection precision.

YOLO-v8 employs a decoupled head and an anchor-free model, facilitating separate handling of the objectness, classification, and regression tasks. This design facilitates focused attention on each task, thereby enhancing the overall accuracy of the model. The anchor-free model can directly predict the object's position and bounding box without the assistance of anchor boxes, Offering the advantage of being more flexible in adapting to objects of different sizes, scales, and shapes. The probability of a bounding box containing an object is indicated by the objectness score in the output layer of YOLO-v8, which is activated using the sigmoid function. The class probabilities, denoting the likelihood of objects affiliated with different classes, are determined using the softmax function.

3.4.2. Loss function

YOLO-v8 uses CIoU (Zheng et al., 2020) and DFL (Li et al., 2020) loss functions for box regression loss and BCE for classification loss.

CIoU (Complete IoU) is a metric that measures the similarity between two bounding boxes, taking into account their position, size, and aspect ratio. By optimizing the CIoU loss, YOLO-v8 aims to minimize the disparity between the predicted bounding boxes and the true boxes, resulting in more precise and tightly fitting detection. The calculation formula for CIoU is as follows:

$$CIoU = 1 - IoU + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \alpha v. \tag{10}$$

where *IoU* represents the traditional Intersection over Union, which measures the overlap between two bounding boxes. c is the Euclidean distance between the center points of the bounding boxes, and ρ is the diagonal length used for distance normalization. v represents the disparity in width-to-height ratios between the two rectangular frame. α is an adjustable hyper parameter in the CIoU calculation, used to balance the importance of position and size.

DFL (Distribution Focal Loss) is a loss function used in object detection tasks to address class imbalance and sample difficulty imbalance. By introducing a distribution focal factor, *DFL* extends the concept of Focal Loss to address the imbalance between positive and negative data. This framework enhances the robustness and detection capabilities of YOLO-v8 by assigning greater weight to challenging instances, enabling the model to focus on arduous samples. The formula for calculating *DFL* can be simply expressed as follows:

$$\mathscr{L}_{DFL} = -\alpha(1 - p_t)^{\gamma} \cdot log(p_t). \tag{11}$$

Here p_t represents the predicted probability of the target class by the model. α is a factor associated with class weights to balance the quantity difference between negative and positive data. γ is an adjustment factor that controls the sensitivity of the focal loss to sample difficulty.

CIoU and *DFL* loss have demonstrated enhanced performance in object detection tasks, particularly when addressing the detection of smaller objects such as marine oil spill.

4. Experimental results

4.1. Generation results

The tools offered by the PyTorch framework are utilized for constructing and training SinGAN models. For SinGAN, the number of pyramid scales is determined by the aspect ratio of the input image. The input image is simultaneously scaled down in both width and height to ensure the shortest side reaches 25 pixels, serving as the coarsest scale input. Each subsequent scale's input image is 1.25 times the size of the previous scale's image, until the longest side of the output image reaches 512 pixels. The weights of all generators and discriminators are updated every 3 iterations. We utilize the Adam algorithm to optimize all the generators and discriminators with a learning rate of 0.0005 and β_1 set to $\frac{1}{2}$. The number of iterations is set to 2000 at each scale. After completing a scale of training, the learning rate will be reduced by the factor of 10. The gradient penalty weight and reconstruction loss weight are set to 0.1 and 10 respectively. After determining the hyper parameters, we individually input all samples containing oil spill into the SinGAN model for training, obtaining the corresponding model parameters for each sample. In order to accelerate the computation speed, we employ GPU to process a large amount of image data, thereby significantly enhancing the efficiency and speed of the training process. Training a single image to completion only requires several minutes.

The number of scales in the pyramid architecture significantly influences the outcomes. When using a smaller number of scales, the model only captures textures. Fig. 10 demonstrates the effect of training with varying numbers of scales. As the number of scales increases, the generators are able to capture larger structures as well as the overall arrangement of oil spill in the scene. Simultaneously, we observed a decreasing trend in the value of SEID (as shown in Fig. 10) as the scale and image resolution increased, from 3.71 to 0.11, suggesting a progressive improvement in image realism. Distorted and low resolution synthetic images often exhibit higher SEID values compared to real samples.

The trained SinGAN can create a significant amount of samples to expand the dataset for neural network training. The original training dataset consists of 620 images with oil spill and 1860 background images. However, relying solely on these original images presents challenges due to insufficient and imbalanced training data, impacting the effectiveness of the training process. By utilizing the generator at a refined scale, we are able to generate a wide range of images featuring diverse oil spill shapes. To ensure quality, we applied filtering based on the SEID values of the generated samples, discarding distorted images, while retaining two to three synthetic samples for each training sample. As a result, the training set is expanded to include a total of 3720 images, with 1860 images containing oil spill. The partial generated samples are illustrated in Fig. 11. Each generated sample maintains the same aspect ratio as the original training samples. The generated samples exhibit novel realistic structures while preserving the overall visual perception. The model is capable of learning different shapes of oil spill and generating oil spill structures with varying textures. We can also observe changes in other objects such as ships and wakes, as depicted in the second row of Fig. 11.

4.2. Data augmentation applied in oil spill detection

The dataset, both prior to and following expansion, is separately employed in training the two YOLO-v8 models with the 20 % of the original images for testing. In the training process, the YOLO-v8 model was initialized with pre-trained weights from the COCO dataset. These pre-trained weights contain valuable knowledge about general object detection tasks learned from a large and diverse dataset. The model was trained using a batch size of 16 for a total of 200 epochs, optimized by SGD with the starting learning rate of 0.015. The set hyperparameters of YOLO-v8 are detailed in Table 2.

The proposed model underwent implementation, training, and validation using data set of two sizes with the help of the visualization tool ClearML. In order to assess the effect of the data augmentation, the dataset both before and after expansion are tested to calculate the results of Recall, Precision and Average Precision (AP). Recall (also known as sensitivity or true positive rate) and Precision are diffusely used metrics for evaluating the effect of detection models, and they focus on different aspects. Recall quantifies the percentage of true positive instances that are correctly identified by the model among all actual positive instances. It emphasizes the model's ability to capture as many true positives as possible, indicating how well the model detects the positive instances. Precision gauges the ratio of veritable positive instances amidst the entirety of samples that the model discerns as true. It focuses on the

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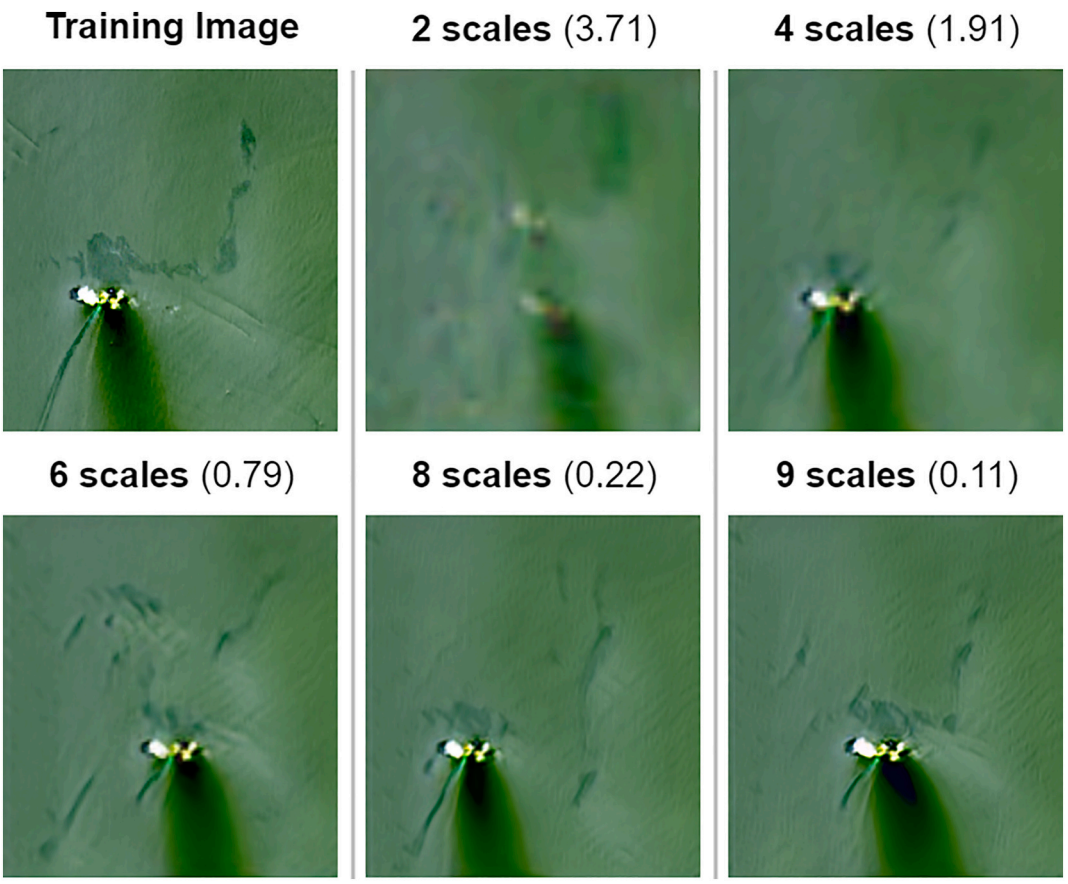


Fig. 10. The effect of training with different scales.

model's ability to make accurate positive predictions, indicating how well the model avoids false positives. Average Precision (AP) takes into account both Recall and Precision by considering the Precision-Recall trade-off. It calculates the zone under the curve of Precision-Recall, which represents the model's detection accuracy across different confidence thresholds. Recall and Precision are calculated using the following formula:

$$Recall = \frac{TP}{FN + TP} \tag{12}$$

$$Precision = \frac{TP}{TP + FP} \tag{13}$$

Here, TP (True Positive) stands for the quantity of true positives correctly identified by the model, FN (False Negative) embodies the quantity of true positives incorrectly classified as negatives, and FP (False Positive) represents the number of negatives incorrectly classified as positives.

Fig. 12(a), (b) illustrates the overall consequences of the model trained on both the unexpanded and expanded balanced dataset, including loss, precision, recall and average precision. It can be seen from Table 3 that the model trained with the unexpanded dataset achieves an average precision (AP) of 71.8 %, while the model trained with the expanded balanced dataset achieves an average precision of 82.0 %. Specifically, we observed a remarkable 11.6 % improvement in Recall and a commendable 5.8 % enhancement in Precision. The use of the expanded balanced dataset for training significantly improves the model's performance.

Fig. 13 shows the outcomes of the proposed model in ocean oil spill detection. By applying the model to the scene of ocean oil spill detection, we have successfully identified and highlighted the presence of oil spill in various maritime settings. Although some of the confidence scores for oil spill detection may be relatively low, our model still demonstrates the capability to identify the rough locations of marine oil spill amidst different backgrounds. This ability is particularly valuable in scenarios where the detection of oil spill is challenging due to complex environmental conditions or varying levels of oil slick visibility. Despite the inherent difficulty in accurately pinpointing every oil spill instance with high confidence, our model provides a valuable initial assessment and aids in directing further investigation and response efforts for marine oil spill pollution.

4.3. Impact of increasing synthetic images on YOLO-v8 model performance

We conduct additional experiments to investigate the impact of increasing the number of synthetic images on the final results of the YOLO-v8 model. Based on the SEID values, we retain more synthetic samples for each real sample and design three experimental groups for analysis. As shown in Table 4, the first group utilizes only background images and generated oil spill images, gradually increasing the quantity of generated images. The second group uses background images and real oil spill images, incrementally increasing the number of background images. The third group employs all background images and real oil spill images, then continuously increasing the quantity of synthesized oil spill images. Group 1 indicates that when using only background images as negative samples without incorporating real oil spill images, the model's performance improves as the number of generated images increases. Group 2 demonstrates that increasing the number of background images slightly enhances the model's performance, primarily in terms of Precision, although the overall improvement is minor. The comparison of group 1 and 2 demonstrates that the models trained with generated images can reach a similar or better detection performance than the

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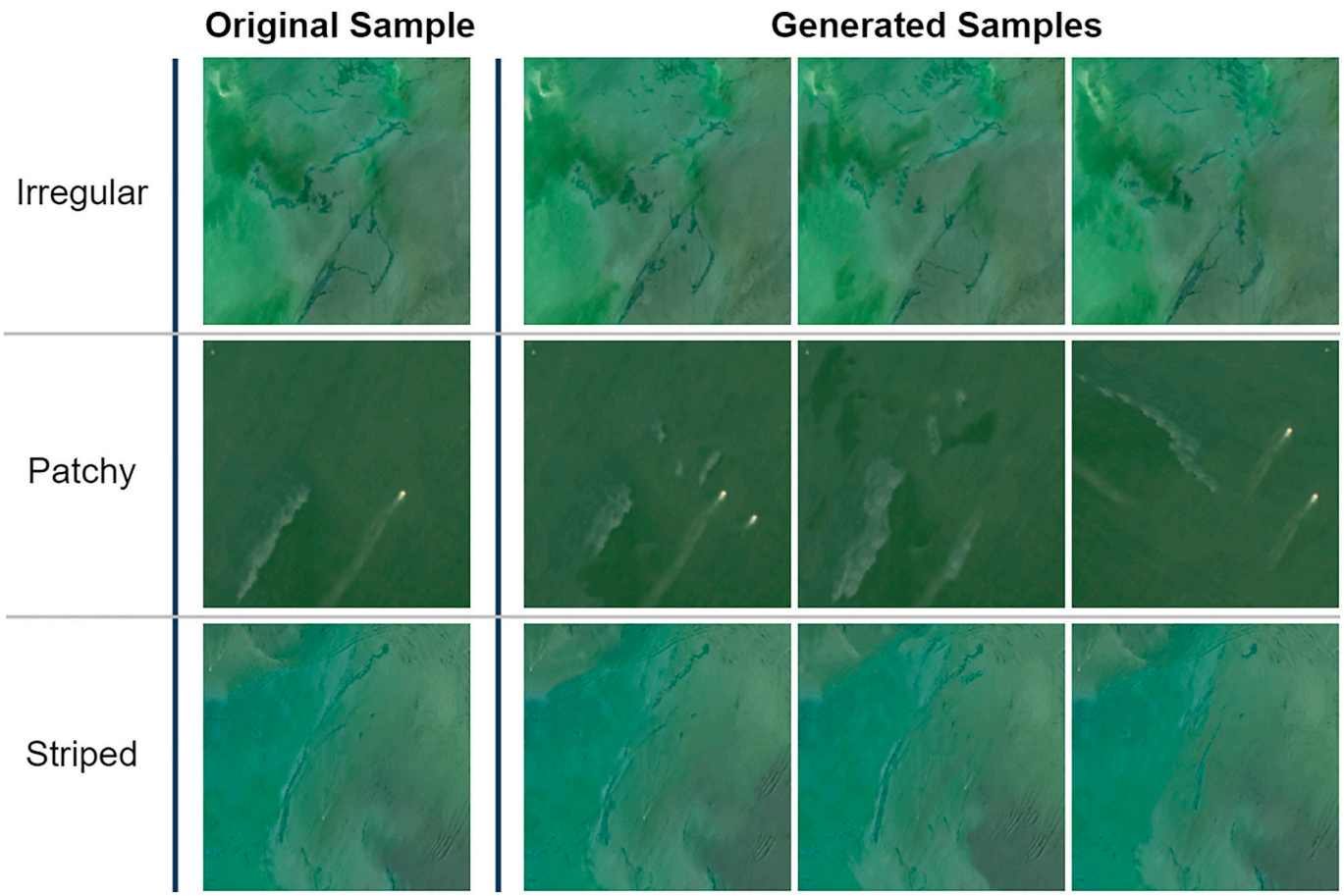


Fig. 11. Generated images using SinGAN. The first image of each row shows the original images.

Table 2
The set hyperparameters of YOLO-v8.

Parameters	Values
Epoch	200
Optimizer	SGD
Image size	640
Batch size	16
Layers	225
Parameters	3,151,904

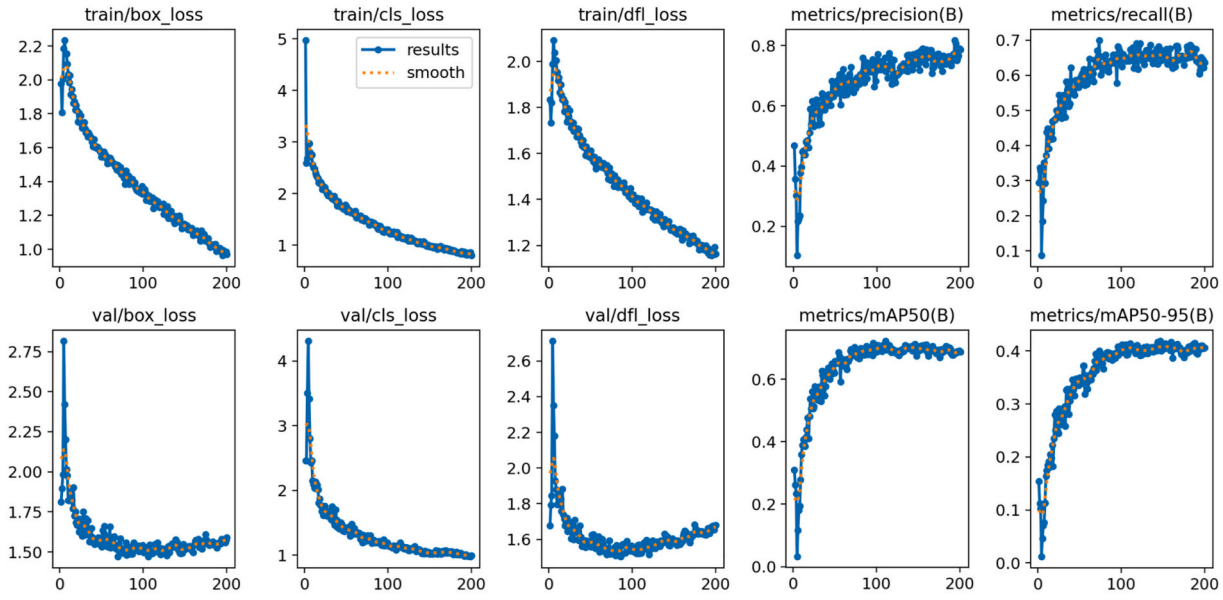
models trained with real images. Group 3 indicates that when using all background images and real oil spill images for training, the model's performance improves with an increasing number of generated images. In theory, we can utilize SinGAN to generate multiple synthetic images corresponding to each real image to enhance the model's performance. However, as the number of generated images increases, the improvement in Recall, Precision, and AP metrics significantly slows down, finally stabilizing at 78.1 %, 80.7 %, and 83.1 %, respectively. The performance improvement of the marine oil spill detection model becomes significantly limited after a certain proportion of generated images, particularly when a balance between background and oil spill images is attained.

4.4. Comparison with other marine oil spill detection models

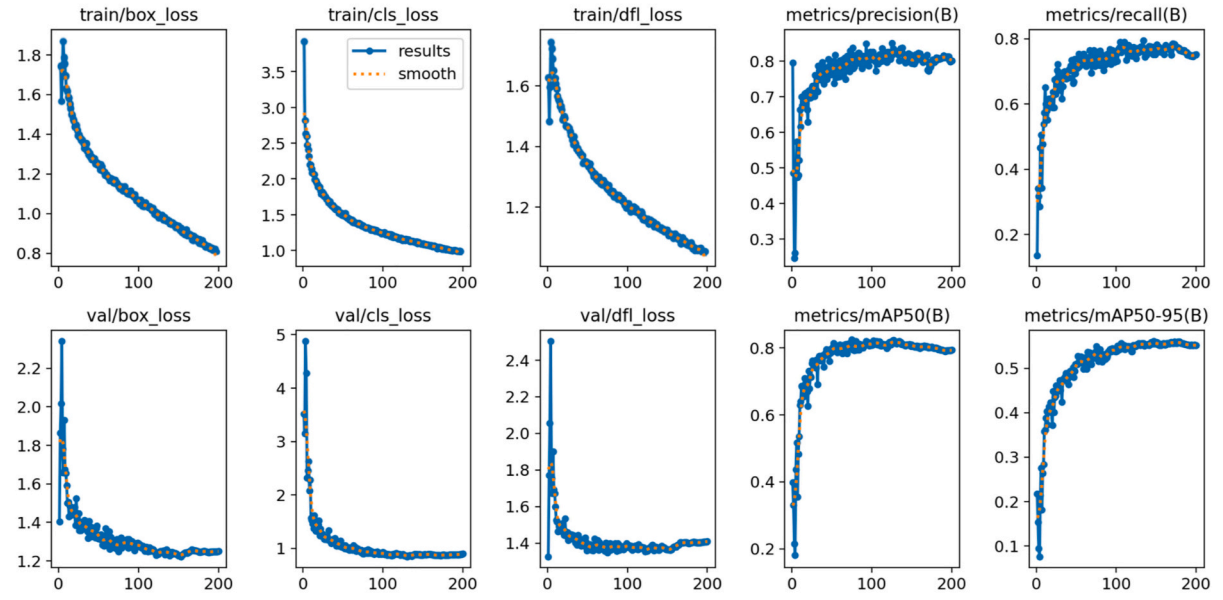
To verify the quantitative effectiveness of the proposed marine oil spill detection method, a comparative study was conducted with several popular deep learning algorithms, including YOLO-v5 (Jocher, 2020), CornerNet (Law and Deng, 2018), Fast R-CNN (Girshick, 2015), and SSD (Liu et al., 2016). The training set both imbalanced and balanced was used to train the oil detection model with the mentioned algorithms, and the test set was used to evaluate the performance of different detection algorithms. The obtained test results are presented in Table 5. It reveals that our marine oil spill detection model based on YOLO-v8 exhibits leading or comparable performance in terms of recall, precision, and AP metrics. It particularly excels in the processing speed of individual images. The single-image prediction time of YOLO-v8 is shorter than other object detection models, with reductions of 70.6 ms, 62.39 ms, 25.06 ms, and 4.04 ms compared to Fast-RCNN, SSD, CornerNet, and YOLO-v5 models, respectively. Additionally, the data augmentation technique based on SinGAN contributes to the performance of other detection algorithms as well.

Fig. 14 provides a more intuitive comparison of the qualitative effects of different marine oil spill detection approaches trained with an expanded dataset. As demonstrated, under good sunglint conditions (such as samples A, D, and H), several marine oil spill detection methods are generally able to detect and outline the location of the spills. However, in some samples (such as E and G), the bounding boxes output by the Fast-RCNN algorithm exceed the appropriate size of the target. The bounding boxes output by the CornerNet algorithm struggle to fully cover the target oil spills (samples B and G). In the presence of partial cloud cover (sample C), SSD and CornerNet algorithms miss small oil spill behind the clouds. YOLO-v5 and YOLO-v8 provide more accurate information about the object's position and size in oil spill detection, with YOLO-v8 achieving a single-image detection time of 8.2 ms, outperforming other marine oil spill detection algorithms and demonstrating stronger competitiveness.

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(a) training with unexpanded datasets



(b) training with expanded datasets

Fig. 12. The overall outcomes of the model trained on both the unexpanded and expanded data set.

Table 3
The detailed results of oil spill detection.

Data set	Ground Truth	Training Samples	Recall	Precision	AP
Dataset before expansion	background	1860	65.8 %	74.4 %	71.8 %
	oil	620			
Balanced dataset after expansion	background	1860	77.4 %	80.2 %	82.0 %
	oil	1860			

5. Conclusions

In this study, we developed a DL-based approach for marine oil spill identification, incorporating data augmentation using SinGAN. By generating diverse synthetic data, we expanded the limited training dataset, meeting the data quantity requirement of the YOLO-v8 model and improving its performance. The use of transfer learning not only expedites the learning process but also reduces the data requirements for the specific task. Compared to other popular detection models, our YOLO-v8-based model not only demonstrates superior or comparable performance in terms of evaluation metrics but also outperforms significantly in terms of the processing speed of individual images.

Furthermore, our method also has some limitations. (1) The manually labeled annotated samples used in our approach may introduce individual errors and a potential presence of a small amount of unreasonable prior information. (2) In specific conditions, such as weak sunlight glare intensity, the YOLO-v8 model may encounter challenges

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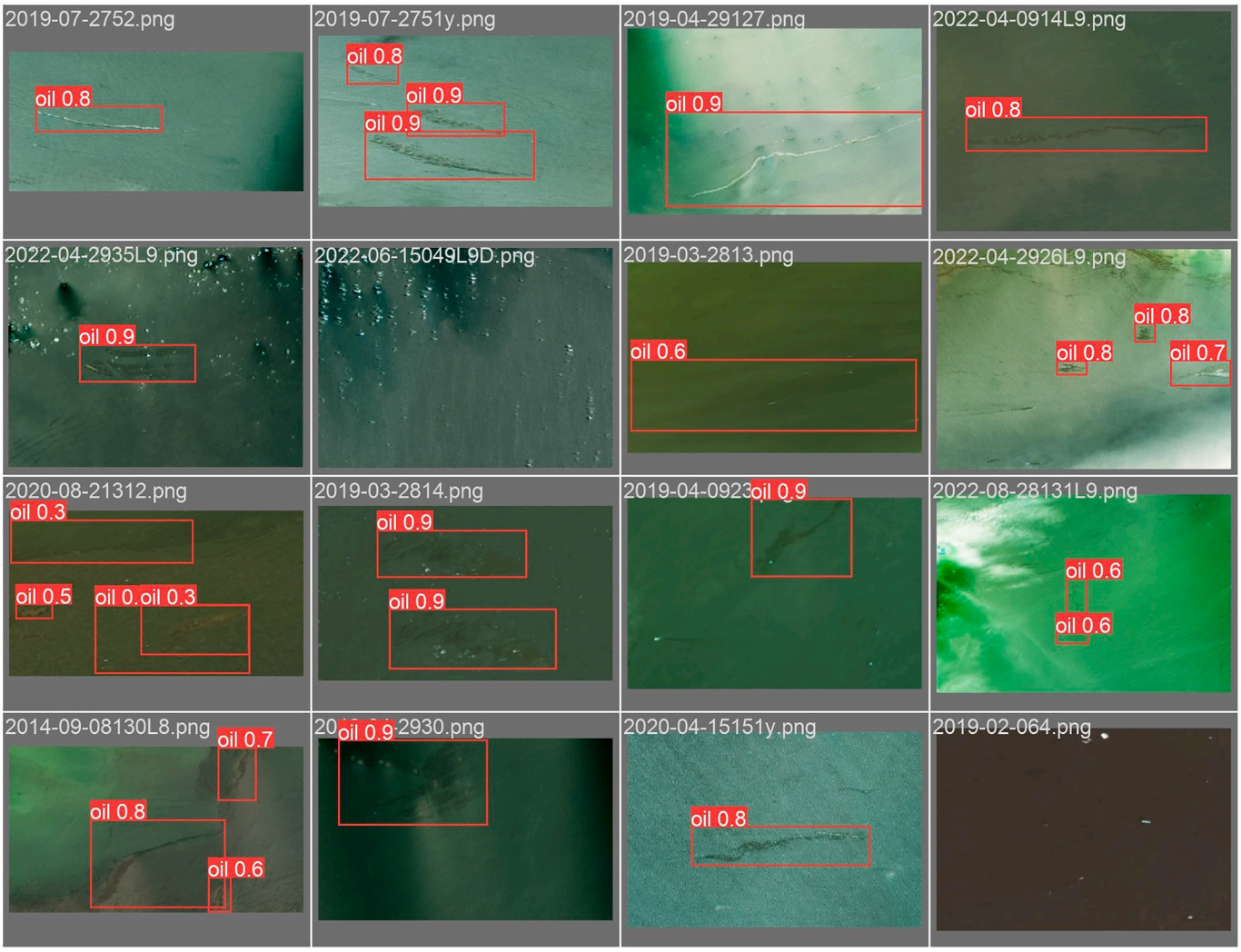


Fig. 13. The predict results of the proposed model in marine oil spill detection.

Table 4
Effect of Varying the Number of Generated Images on Performance.

Groups	Real background image	Oil spill image		Recall	Precision	AP
		Real	Generated			
1	1860	0	620	57.8 %	62.8 %	62.1 %
			620 × 2	62.6 %	68.4 %	67.6 %
			620 × 3	64.7 %	70.3 %	69.8 %
2	0	620	0	64.9 %	70.1 %	68.4 %
	620			65.6 %	73.8 %	71.6 %
	620 × 2			65.7 %	74.2 %	71.7 %
	620 × 3			65.8 %	74.4 %	71.8 %
3	1860	620	620	74.0 %	77.1 %	78.9 %
			620 × 2	77.4 %	80.2 %	82.0 %
			620 × 3	77.8 %	80.6 %	82.7 %
			620 × 4	78.0 %	80.7 %	83.0 %
			620 × 5	78.1 %	80.7 %	83.1 %

in accurately detecting oil spill. How to accurately detect marine oil spill under complex environmental conditions is our next research direction.

Future research can further improve the model's robustness and generalization in complex sea surface environments. Exploring additional data augmentation techniques and optimization algorithms also holds promise. We believe that this study provides a valuable foundation and insights for the advancement of marine oil spill monitoring and environmental protection.

CRediT authorship contribution statement

Yuepeng Cai: Writing – original draft, Investigation, Formal analysis, Data curation. **Lusheng Chen:** Writing – review & editing, Validation, Software, Data curation. **Xuebin Zhuang:** Writing – review & editing, Supervision, Formal analysis. **Bolin Zhang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 5
Quantitative comparative study of the proposed method with existing methods.

	Backbone	Dataset	Recall	Precision	AP	Test time per image (ms)
Fast-RCNN	VGG16	Imbalanced	66.9 %	66.2 %	68.5 %	78.80
		Balanced	72.3 %	72.7 %	75.6 %	
SSD	VGG16	Imbalanced	64.2 %	65.9 %	65.5 %	70.59
		Balanced	69.6 %	74.6 %	74.8 %	
CornerNet	Resnet50	Imbalanced	60.8 %	71.3 %	69.9 %	33.26
		Balanced	67.2 %	73.6 %	77.6 %	
YOLO-v5	CSPDarknet53	Imbalanced	64.6 %	73.7 %	71.2 %	12.24
		Balanced	77.6 %	79.8 %	81.7 %	
YOLO-v8	C2F module	Imbalanced	65.8 %	74.4 %	71.8 %	**8.2**
		Balanced	**77.4 %**	**80.2 %**	**82.0 %**	

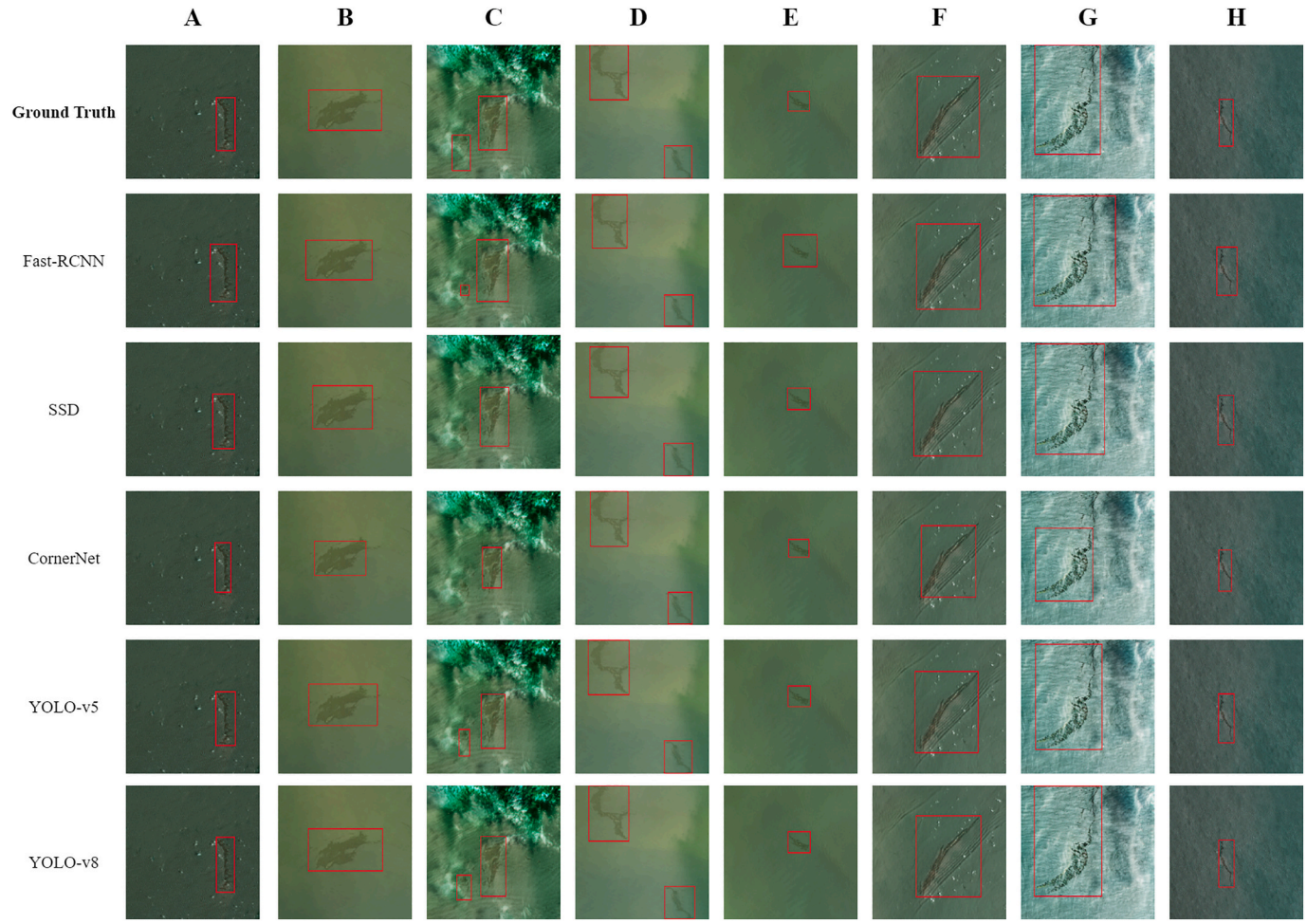


Fig. 14. Qualitative comparison between the proposed method and existing methods.

Data availability

Data will be made available on request.

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